```
#include <iostream> using namespace std;

int binarySearch(int A[], int v, int p, int r) {
    if (p > r) {
        return -1;
    }

    int j = p + (r - p) / 2;

    if (v == A[j]) {
        return j;
    } else if (v < A[j]) {
        return binarySearch(A, v, p, j - 1);
    } else {
        return binarySearch(A, v, j + 1, r);
    } }

int main() {
    int A[] = {2, 5, 8, 12, 16, 23, 38, 56, 72, 91};
    int n = sizeof(A) / sizeof(A[0]);

    cout << "Sorted array: ";
    for (int i = 0; i < n; i++) {
        cout << A[i] << " ";
    }
    cout << endl;

    int searchValues[] = {23, 50, 2, 91, 10};
    int numSearches = sizeof(searchValues) / sizeof(searchValues[0]);

    for (int i = 0; i < numSearches; i++) {
        int v = searchValues[i];
        int result = binarySearch(A, v, 0, n - 1);
        if (result != -1) {
            cout << "Value " << v << " found at index " << result << endl;
        } else {
            cout << "Value " << v << " not found in the array" << endl;
        }
    }

    return 0; }
```